AquaSense

The purifier that knows your water and who drinks it.

Add an edge-first sensing and identity layer so every dispense becomes a measured, attributable, and actionable event.

Member Name	Role & Subsystem
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Targets are engineering design targets to be validated in prototyping.

SECTION B • PROBLEM & IDEA FRAMING & SOLUTION MECHANISM

Why Purifier Health & Hydration Belong Together

AREA OF EXPLORATION

Dual-Track Integration

IoT Device Monitoring: Live TDS, pH, filter life, remote flush.

Personalised Hydration: Automated per-member intake tracking.

Core Insight: Intake tracking requires verified water quality.

Connected Purifiers Leave Users Guessing

- **Blind Water Quality:** Zero visibility into live TDS or pH.
- **Calendar Filter Changes:** Fixed timers ignore actual water usage.
- **Manual App Logging:** Users abandon manual hydration apps quickly.
- **Physical Maintenance:** Draining stagnant tanks requires physical presence.

How Every Dispense Becomes an Actionable Event

STEP 01

Flow & Burst

Flow triggers 3–5 camera frames.

STEP 02

On-Device ID

Pi matches local face embeddings.

STEP 03

Edge Log

Volume, TDS, temp to SQLite.

STEP 04

MQTT Sync

TLS upload to cloud dashboard.

STEP 05

Remote Care

App triggers flush with ACK.

SECTION B • DESIGN TARGETS TO VALIDATE & TECHNICAL DETAILS

Baseline vs. Target Performance Benchmarks

Parameter	Baseline (Typical Purifier)	Our Design Target (Prototyping Validation)
Per-user intake tracking	None / manual logging	**Automatic, ≥ 90% indoor ID accuracy**
Dispensed volume accuracy	Not measured	**± 5% after flow-sensor jug calibration**
Filter replacement trigger	Fixed calendar / guess	**Usage alert @ 90% life (litres + TDS-in/out)**
Remote maintenance	Physical presence required	**Executed within ≤ 5 s when online with ACK**
Water-quality visibility	None	**Live TDS (± 10% calibrated) & indicative pH**
Wi-Fi outage data loss	N/A	**0 events lost via local SQLite buffering**

Privacy by Design & Reliable Sync

- On-device embeddings only with physical camera shutter.
- Deduplicated event IDs with Firebase urgent alerts.

SECTION C • TECHNICAL FEASIBILITY

System-Level Block Diagram (Edge → Cloud → App)

1. PURIFIER EDGE LAYER (LOCAL AUTHORITY)

Sensors & Vision Input

- YF-S201 Flow
- Dual TDS + pH + DS18B20 (ADS1115)
- PIR + Pi Camera

Raspberry Pi 4B / Zero 2 W

- MobileFaceNet (ONNX/TFLite)
- Local SQLite Buffer
- Interlocks

Isolated Actuation

Opto-Isolated Relay / MOSFET → 12 V Food-Grade Drain Valve

MQTT / TLS

↔

Events ↑ Cmds ↓

2. CLOUD BACKEND

Broker & API

Mosquitto MQTT → FastAPI / Node

Storage & Alerts

PostgreSQL + Firebase Push

REST API

↔

3. COMPANION APP

Flutter / React Native

Personal daily intake, live TDS/pH, filter life, remote drain/flush.

MobileFaceNet Vision (< 1 s)
ONNX/TFLite 128–512d cosine match with blink liveness check.

Calibrated Flow & Proximity Wake
YF-S201 (~450 pulses/L) + 1 m ultrasonic/PIR wake trigger.

Compensated Dual-TDS & pH
ADS1115 16-bit I²C ADC + DS18B20 temperature compensation.

Offline-First SQLite & Outbound TLS
Local write-ahead buffer + NAT-friendly outbound MQTT over TLS.

SECTION D • SAFETY CRITICAL ASPECTS & EXECUTION FEASIBILITY

Safety Authority Remains Strictly at the Edge

- **Low-Voltage Isolation:** 5 V / 12 V DC sensing; opto-isolated relay/MOSFET drivers with flyback protection.
- **Food-Grade & Dry Enclosure:** Sealed electronics enclosure away from wet zone; food-grade wetted parts.
- **Firmware Interlocks:** Blocks remote drain/flush commands during active water dispensing.
- **Dry-Run & Leak Shut-Off:** Tank float/level sensor prevents running pumps dry.
- **Additive Connectivity:** Admin TLS auth logged, normal safe dispensing continues offline.

Component List, Costing & Sourcing Mitigations

Prototype Component List

- **Compute & Vision:** Pi 4B / Zero 2 W, Pi Camera + IR/LED ring.
- **Flow & Quality:** YF-S201 flow, TDS ×2, pH probe, DS18B20 temp.
- **ADC & Level:** ADS1115 16-bit ADC, tank float/level sensor.
- **Actuation & Wake:** 12 V food-grade solenoid + relay, PIR sensor.
- **Hardware UI:** LED ring, privacy shutter, 3D enclosure, optional BLE/NFC.

Order-of-Magnitude Cost

Prototype Add-On: ≈ ₹10,000 – 15,000 (Pi, camera, pH probe).

At Volume: ≈ ₹3,000–4,500/unit (+10–15% on mid-range purifier).

Sourcing Risks & Mitigations

- **Pi/Camera Stock:** Early order via authorised distributors.
- **pH Drift (1–2 wk lead):** Treated as indicative with calibration.
- **Solenoid & 3D Mount:** Verify food-grade spec; 1–3 day 3D print.

4-Week Build Plan to Level 1 Prototype

- **Week 1:** Order parts (pH probe and valve first). Set up the Pi, server and app login. Calibrate the flow sensor with a measuring jug.
- **Week 2:** Add TDS and temperature sensors. Build camera capture on dispense and face matching on the Pi, with local logging.
- **Week 3:** Sync data to the server and app. Add the drain command (admin only, blocked during dispensing) and one TDS/filter alert.
- **Week 4:** Assemble, test, record a demo.